Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 (Generated practice paper - Set 15)

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

Do not write outside the box

**Answer all questions in the spaces provided.**

**0 1** Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

**Table 1**

| **Number of nights** | **1** | **2** | **4** | **6** | **8** | **10** | **12** | **14** |
|---|---|---|---|---|---|---|---|---|
| Running mean catch per night | 31 | 18 | 24 | 21 | 22 | 22 | 21 | 22 |

**0 1 . 1** Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

**[2 marks]**

.....

.....

.....

.....

**0 1 . 2** Explain why moth surveys must be carried out at standardised times and weather conditions.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

**0 1 . 3** Distinguish between **accuracy** and **precision** of measurements.

**[2 marks]**

.....

.....

.....

.....

Do not write outside the box

**0 1 . 4** Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

**10**

**0 2 . 1** Describe the main features of the Montreal Protocol (1987).

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

**0 2 . 2** For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

**[4 marks]**

Do not write outside the box

**Aerosol propellant** .....

.....

.....

.....

**Refrigerant** .....

.....

.....

.....

**0 2 . 3** Describe how waste CFCs from old refrigerators are disposed of safely.

**[2 marks]**

.....

.....

.....

.....

**0 2 . 4** State **one** reason why the ozone layer is not expected to recover fully for many decades.

**[1 mark]**

.....

.....

**10**

**0 3 . 1** Explain the locational requirements for harnessing wave power.

**[2 marks]**

.....

.....

.....

.....

**0 3 . 2** Describe how **one** named wave power device generates electricity.

**[2 marks]**

.....

.....

Do not write outside the box

Do not write outside the box

.....

.....

**0 3 . 3** Give **two** design problems that have delayed the development of wave power.

**[2 marks]**

**1** .....

.....

**2** .....

.....

**0 3 . 4** Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

**[4 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

**10**

**0 4 . 1** Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

**[1 mark]**

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

**0 4 . 2** Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

**[3 marks]**

Do not write outside the box

**0 4 . 3** Explain why crumb peds produce more fertile soil than platy peds.

**[1 mark]**

**5**

**0 5 . 1** Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

**[3 marks]**

**0 5 . 2** Explain how mycorrhizal fungi increase phosphate uptake by plants.

**[2 marks]**

**0 5 . 3** Describe how human activities have altered the phosphorus cycle.

**[3 marks]**

Do not write outside the box

**0 5 . 4** Explain **two** strategies for the sustainable management of phosphorus supplies.

**[2 marks]**

**10**

**0 6** The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

**Table 1**

| **Year** | **1960** | **1980** | **2000** | **2015** |
|---|---|---|---|---|
| Lake area / $km^2$ | 68 000 | 52 000 | 24 000 | 7 000 |
| Salinity / g $dm^{-3}$ | 10 | 18 | 45 | over 100 |
| Commercial fish catch / thousand tonnes | 40 | 10 | 0 | 0 |

**0 6 . 1** Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

**[2 marks]**

Answer [ ] %

**0 6 . 2** Explain why the salinity of the lake increased as the area fell.

**[2 marks]**

.....

.....

.....

.....

**0 6 . 3** Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

**[4 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

**0 6 . 4** Suggest why disagreements over the rivers continue between the countries of the region.

**[2 marks]**

.....

.....

.....

.....

Do not write outside the box

10

**0 7 . 1** Give **three** sources of oil pollution of the sea other than tanker accidents.

**[3 marks]**

**1** .....

.....

**2** .....

.....

**3** .....

.....

**0 7 . 2** Explain how an oil spill at sea can kill seabirds.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

**0 7 . 3** Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

**[2 marks]**

.....

.....

.....

.....

**0 7 . 4** Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

**[2 marks]**

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

10

**0 8** Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

**Table 1**

| Condition | Mean surface temperature / °C |
|---|---|
| No greenhouse effect | -18 |
| Natural greenhouse effect | 15 |

**0 8 . 1** Use Table 1 to calculate the warming produced by the natural greenhouse effect.

**[1 mark]**

**Answer** ☐ °C

**0 8 . 2** Explain how the natural greenhouse effect warms the Earth's surface.

**[4 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

**0 8 . 3** Explain what is meant by a **greenhouse gas**.

**[1 mark]**

.....

.....

**0 8 . 4** The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

**[2 marks]**

.....

.....

.....

.....

**0 8 . 5** Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

**[2 marks]**

.....

.....

.....

.....

**10**

**0 9** A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

**0 9 . 1** Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

**[2 marks]**

.....

.....

.....

.....

**0 9 . 2** Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

**[4 marks]**

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

**0 9 . 3** Explain how the results of the analysis reduce the cost of environmental monitoring.

**[2 marks]**

.....

.....

.....

.....

**0 9 . 4** Explain why Critical Group Monitoring measures exposure rather than damage to health.

**[2 marks]**

.....

.....

.....

.....

**10**

Do not write outside the box

**1 0 . 1** Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

**[9 marks]**

Do not write outside the box

Do not write outside the box

Extra space

**1 0 . 2** State what is meant by **bioaccumulation**.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

**1 1 . 1** To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

**[25 marks]**

**OR**

**1 1 . 2** Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

**END OF QUESTIONS**